UMass Amherst

Info Fusion - Sage Lab Synthesis:

Improving antibiotic resistance prediction with mutli-modal input

Jackson Michaels

Project Outline

Background - Bio

Genetic Vocab:

- DNA strand of **base pairs** aka **nucleotide pairs**
- sets of three **(codons)** are encoded as **amino acids**
- amino acids come together to construct **proteins**
- generally, **one protein per gene** for TB
- Regions can be **genic** (inside a gene) or **intergenic** (outside a gene)
    - the latter are of particular interest because they regulate and promote gene expression

Project Vocab:

- previous works have identified sets of relevant regions in the genome referred to as **Loci**
- **Scope,** A locus can be entirely genic, entirely intergenic, or span across both.

Background - Project

Goal: take existing work from SAGE lab on antibiotic resistance prediction in TB and improve on performance and interpretability by incorporating multimodal insights from Info Fusion lab

Task Details:

- Original input data is presented as set of loci, 5 X ~7,500 vector of base pairs for single drug
  - these loci can be genic regions or intergenic
  - typically one hot encoded *(A, C, T, G, gap)*
  - covers 2-3 genes and intergenic regions
- task is to predict resistance to eleven antibiotic drugs
  - multi-drug input is concatenated single-drug inputs

Modalities - DNA

Top Choices:

- Regulatory regions
- more than 2-3 proteins

DNA-Derived:

- one-hot array / ref-alt encoding
- foundation model embedding
- K-mer frequency (31-mers used in literature)

```
>NC_000962.3 NC_000962.3:314309..314854 (-
strand) class=CDS length=546

GTGCACACCCAGGTACACACGGCCCGCCTGGTCCAC
ACCGCCGATCTTGACAGCGAGACCCGCCAGGACATC
CGTCAGATGGTCACCGGCGCGTTTGCCGGTGACTTCA
CCGAGACCGACTGGGAGCACACGCTGGGTGGGATGC
ACGCCCTGATCTGGCATCACGGGGCGATCATCGCGC
ATGCCGCGGTGATCCAGCGGCGACTGATCTACCGCG
GCAACGCGCTGCGCTGCGGGTACGTCGAAGGCGTTG
CGGTGCGGGCGGACTGGCGGGGCCAACGCCTGGTG
AGCGCGCTGTTGGACGCCGTCGAGCAGGTGATGCGC
GGCGCTTACCAGCTCGGAGCGCTCAGTTCCTCGGCG
CGGGCCCGCAGACTGTACGCCTCACGCGGCTGGCTG
CCCTGGCACGGCCCGACATCGGTACTGGCACCAACC
GGTCCAGTCCGTACACCCGATGACGACGGAACGGTG
TTCGTCCTGCCCATCGACATCAGCCTGGACACCTCGG
CGGAGCTGATGTGCGATTGGCGCGCGGGCGACGTCT
GGTAA
```

Example Loci: Rv0262c

Modalities - Regulatory

Regulatory Regions:

- high-value, mechanistically-motivated modality that current CNN baselines under-use
- handled as separate modality from DNA

Description:

- regulates the downstream gene(s) it precedes
  - e.g. promoters / operators
- regulates expression or lack of expression for previous genes

Motivation:

- Resistance can arise without changing the protein at all, by a promoter mutation that over-expresses the gene

Modalities - Protein

Description:

- Translation from DNA sequence
- effectively 3-mers

Representations:

- one-hot array / ref-alt encoding of amino-acids
- foundation model embedding

Notes:

- another top choice is to simply include more than 2-3 in the inference

```
>Mycobacterium tuberculosis

H37Rv|Rv0262c|aac

VHTQVHTARLVHTADLDSETRQDIRQMVTGAFAGDFT
ETDWEHTLGGMHALIWHHGAIIAHAAVIQRRLIYRGNA
LRCGYVEGVAVRADWRGQRLVSALLDAVEQVMRGAYQ
LGALSSSARARRLYASRGWLPWHGPTSVLAPTGPVRTP
DDDGTVFVLPIDISLDTSAELMCDWRAGDVW
```

Example Loci: Rv0262c, amino-acid sequence

Modalities - 3D Structure

Description:

- spatially describe distance of base-pairs in folded protein

Representations:

- distance matrix (e.g. alpha fold)
- Density vector (mine)
- Shrake-Rupley algorithm

Notes:

- only useful if two proteins interact

Modalities - Molecular Characteristics

Description:

- Chemical properties of molecules in protein separate from function
- Represents how individual molecules behave and interact
- Used in *(Kulkarni et al)*

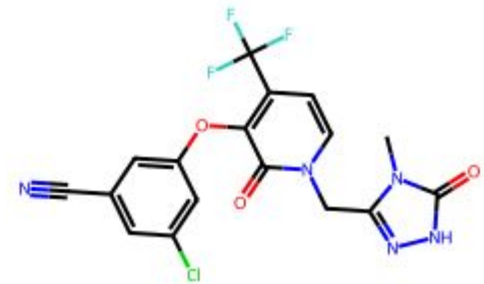
Features:

- Molecular weight
- Isoelectric point
- Hydro / Lipophilicity
- Refractivity

Notes:

- derived from RDKit Library

RDKit
Open-Source Cheminformatics and Machine Learning



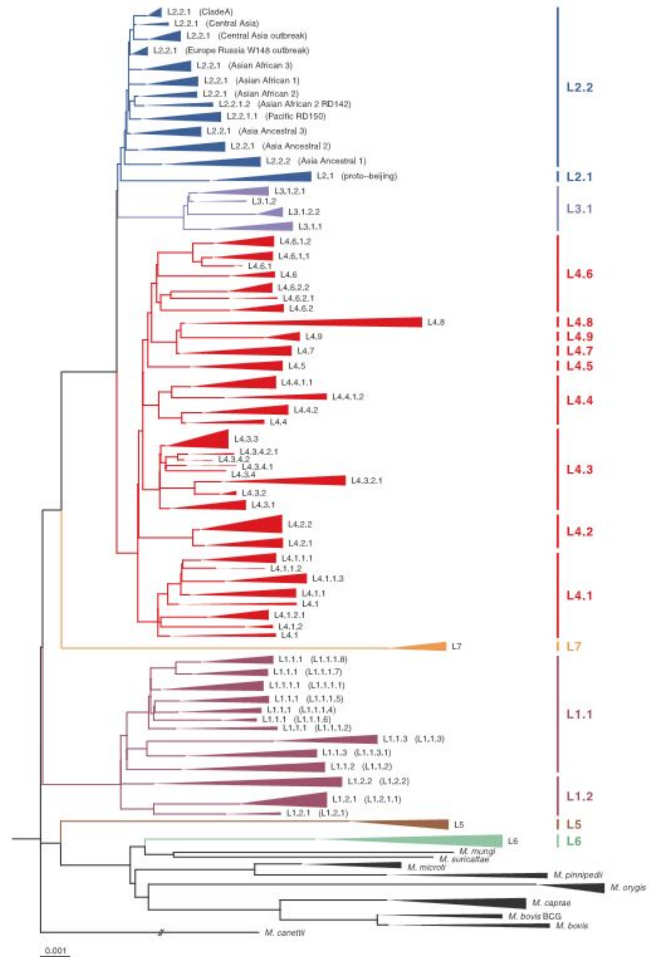

```
getMolDescriptors(doravirine)

{'MaxEStateIndex': 13.412553309006833,
 'MinEStateIndex': -4.871620672188628,
 'MaxAbsEStateIndex': 13.412553309006833,
 'MinAbsEStateIndex': 0.045220418860841605,
 'qed': 0.6914051268589834,
 'MolWt': 425.754,
...
'fr_thiophene': 0,
 'fr_unbrch_alkane': 0,
 'fr_urea': 0}
```

Modalities - Lineage

Description:
- evolutionary lineage of specific TB strain

Representation:
- 1D binary vector, N X N matrix, tree structure

Notes:
- Literature is split on using lineage as its commonly a confounder
- I will ablate with and without lineage

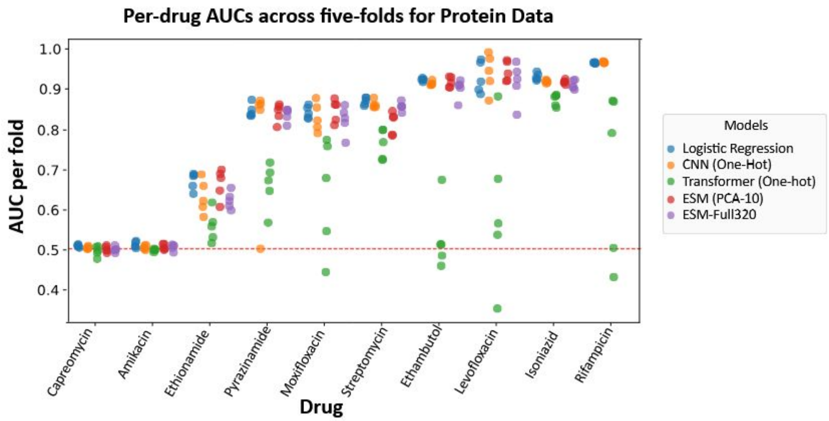


Model Design

Models - Literature

Trends:

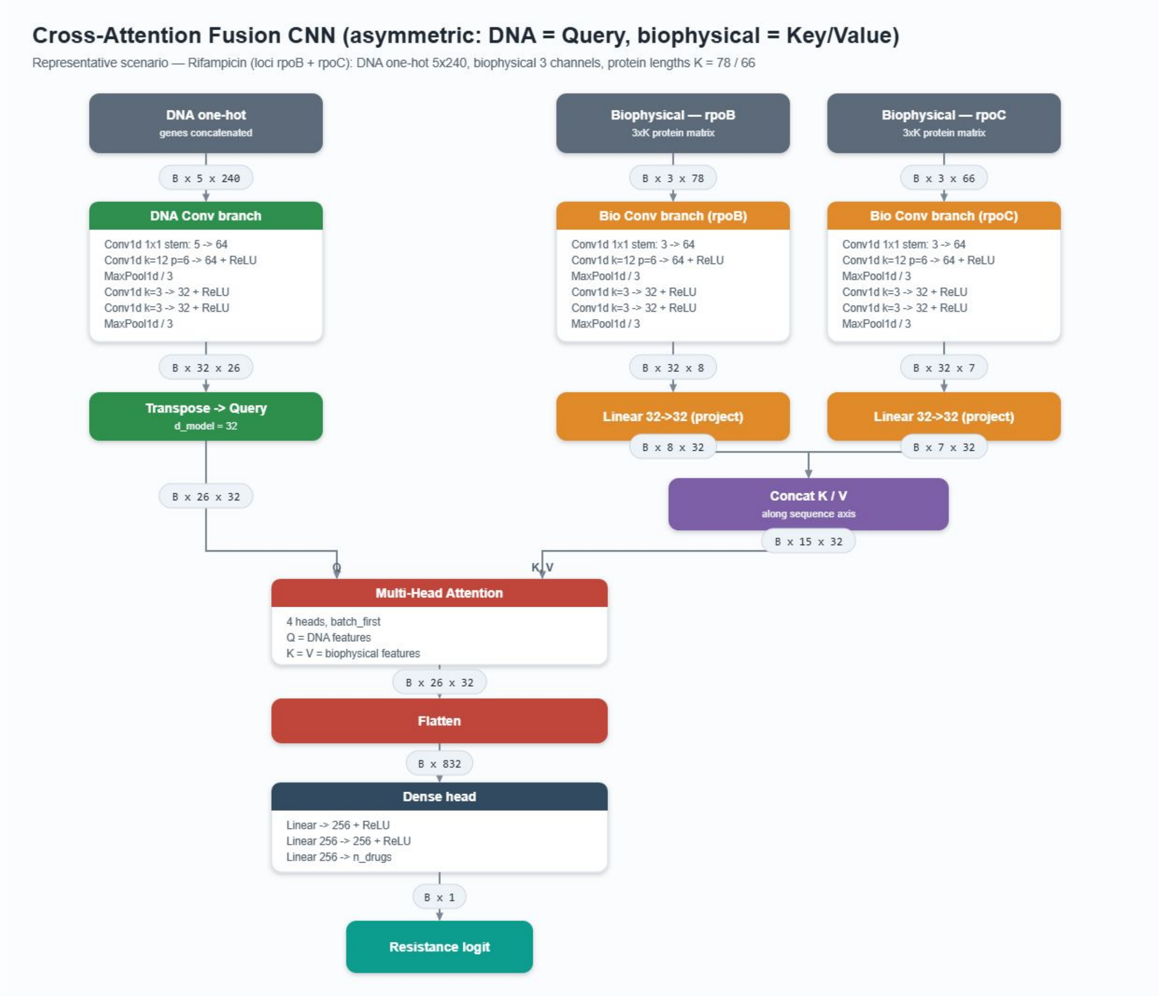
- CNN models consistently outperform other architectures
  - particularly on DNA sequence tasks
- DNA-sequence models consistently outperform protein-based models, likely because they capture crucial non-coding regulatory regions.
- Despite their complexity, language models struggle with second-line drugs and sequence-position averaging, largely due to data bottlenecks.



Models - Cross Attention

Description:

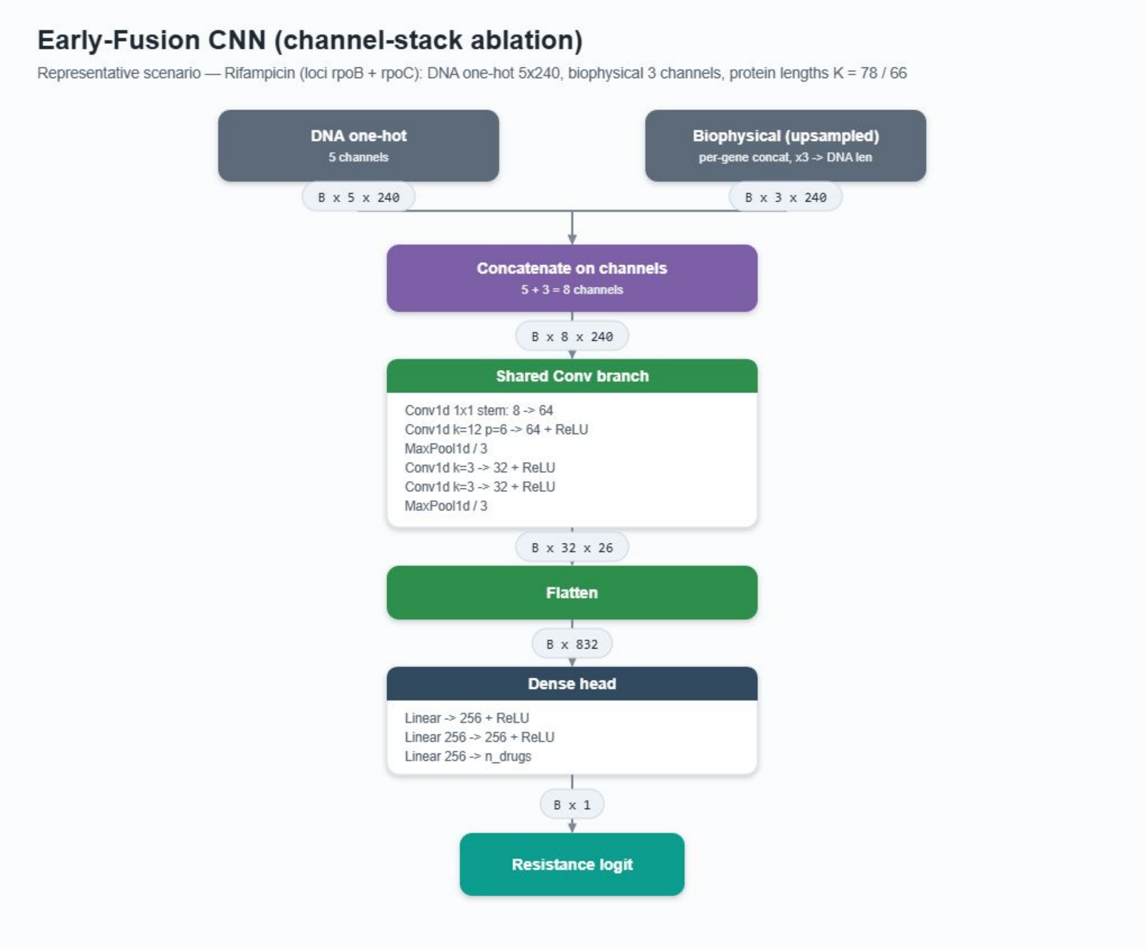
- to retain 1D CNN performance the model uses cross attention treating DNA as the query and other modalities as the key / value
- Should allow CNN insights to guide inference only retrieving information as needed



Models - Early Fusion

Description:

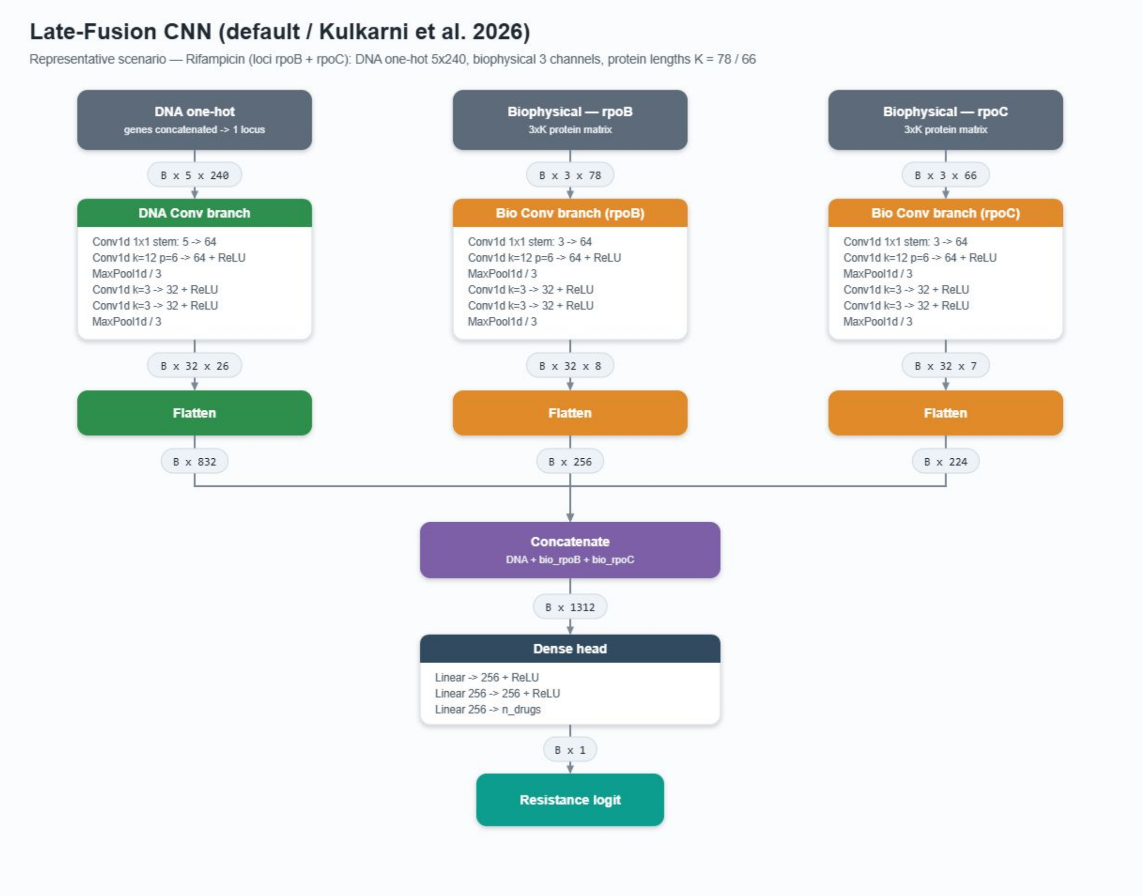
- concatenate all modalities into single vector
- pass concat vector into model as normal
- Fewer parameters and simpler implementation but lacking multi-modal nuance



Models - Late Fusion

Description:

- separate stacks for each modality
- allows different models to process different modalities
    - e.g. CNN for DNA and transformer for 3D distance
- requires more parameters but more robust
- can be trained with modality drop-out



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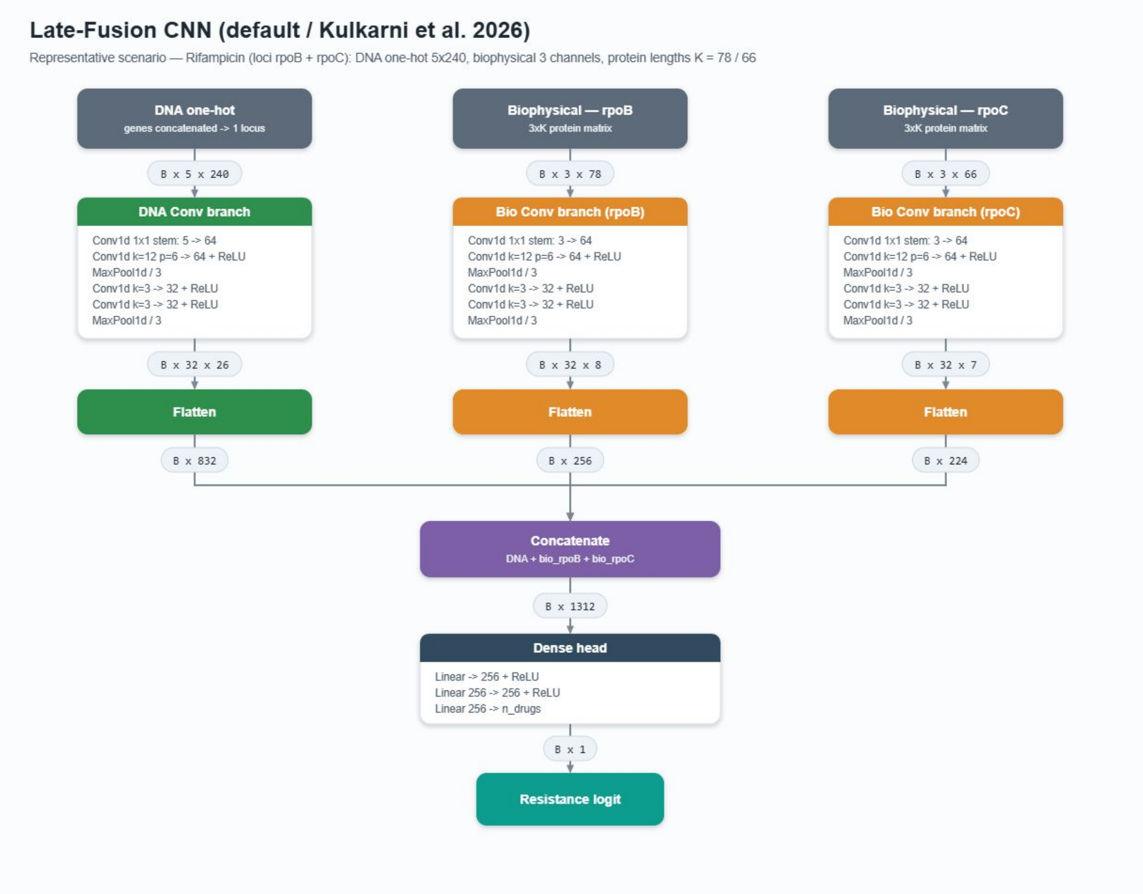

Interpretability

Interp - SHAP

Description:

- Open Source
- mechanistic interpretability library for input attribution
- commonly used in Comp Bio papers